

Vaibhav Chaudhari

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EDUCATION

New York University Courant

New York, NY

Master of Science in Computer Science GPA: 3.78

September 2025 – May 2027(Expected)

BITS Pilani K.K. Birla Goa Campus

Goa, India

Bachelor of Engineering in Computer Science and Master of Science in Physics

August 2017 – June 2022

TECHNICAL SKILLS

Languages: Go (Golang), Python, C, C++

Systems & Infrastructure: gRPC, Protocol Buffers, REST APIs, Linux, Multi-threading

Tooling & Observability: Jaeger, CI/CD (Jenkins), Git, Postman, Grafana, Elasticsearch

Machine Learning: PyTorch, Transformers, DPO, NLTK, spaCy, OpenCV, Pandas, NumPy

EXPERIENCE

Graduate Research Assistant, NYU (advised by Dr. Dennis Shasha)

May 2026 – Present

- Engineered **VecFast** – a **High-Performance Concurrent Replicated Vector Database** on **HNSW graphs**, sustaining baseline recall (**Recall@10 = 0.996**) under concurrent read/write load.
- Designed a **root-branching partitioned index** that shards vector space via k-means routing with a soft-routing trick, achieving up to **14.75× write throughput** over a monolithic index at **K=8** while holding **99.1%** of baseline recall.
- Developed a **concurrent mixed-workload benchmark** proving VecFast maintains full query QPS under simultaneous insert load, where single-node systems (hnsplib, FAISS, ScaNN) stall.
- Built a **write-all/read-one replication layer** broadcasting inserts to N replicas with round-robin reads, yielding near-linear read-QPS scaling at unchanged recall, plus **hot-shard cloning** to cut P99 tail latency under skewed traffic.

NeuroMap-MS | Research Assistant, NYU (advised by Dr. Artie Shen)

June 2026 – Present

- Building an automated MRI-to-biomarker pipeline converting routine brain and spinal cord MRI into a whole-neuraxis lesion topography profile, linking lesion location to disability (EDSS) and progression in multiple sclerosis.
- Own the **spinal cord arm** of the pipeline, covering spinal lesion segmentation, preprocessing, and quality control for routine clinical MRI.
- Benchmark and evaluate public datasets and state-of-the-art **spinal cord lesion segmentation** methods; run candidate pipelines and document requirements, limitations, and test outputs.
- Develop a **cervical vs. thoracic localization strategy** to map lesion masks to vertebral/spinal levels, producing region-specific lesion burden features (count, volume, % involvement).
- Contribute spine-derived features to a unified imaging-clinical dataset used to test whether topographic lesion burden adds predictive value over conventional total brain lesion volume.

Nutanix

July 2022 – July 2025

Software Engineer II (MTS-3), R&D-AHV

Bangalore, India

- Built the flagship **v4 VMM (Virtual Machine Management)** control plane for AHV and ESXi hypervisors in **Go, Python, gRPC, and Protocol Buffers**, with **p99 response time under 200ms**.
- Engineered v4 infrastructure features including **Selection Projection, HATEOAS, RBAC governance, IAM pipelines**, and complex filtering for NIC Subnet UUIDs and GPU configurations.
- Reduced API latency by **30%** using **Jaeger distributed tracing** to refactor sequential calls for Go concurrency and implement a **TTL IAM cache**.
- Implemented **DODIN compliance** and **Flashmode** features for secure VM lifecycle management.
- Resolved critical **on-call escalations** involving Acropolis crash loops and VM power-on failures by purging scheduler segments and resolving parcel corruption.
- Mentored interns on benchmarking and performance improvement projects for ESXi and AHV v4 GET/LIST APIs.
- Internship (Fall 2021):* Automated Uhura RPC benchmarking using **Go, Python, Grafana, and Elasticsearch**.

University of Eastern Finland

January 2022 – June 2022

Research Assistant

Joensuu, Finland

- Conducted Part-of-Speech tagging and Named Entity Recognition on **2.7M+ tweets** from Nordic countries using **Python, NLTK, and spaCy**.
- Evaluated NLP tools on multilingual social media text, producing insights, visualizations, and statistical reports on linguistic and entity distribution in Nordic Twitter datasets.

ServiceNow

May 2021 – July 2021

Software Development Intern

Hyderabad, India

- Built an **Engagement side panel** and multiple **Overview Pages** within the ServiceNow platform using **UI Builder** and **JavaScript**, improving operator workflow visibility across service management modules.

PROJECTS

DownShift - Self-Healing LLM Analytics Compiler | Github Link

September 2026

- Architected a **JIT-compiler-style system** on **MongoDB Atlas** and **OpenRouter** that converts a frontier-model answer into a reusable, tested **skill**, letting an 8B model answer similar questions at up to **200× lower cost**.
- Achieved **100% accuracy (30/30)** on held-out benchmark questions at \$0.0031/question vs. \$0.00408 for frontier-only, driving steady-state cost to **\$0.00002/answer** at **< 1s** median latency once a skill is learned.
- Designed deterministic **guardrails** (typed parameter validation, keyword grounding, calendar-aware date-range checks, read-only pipeline enforcement) around the cheap-model executor to block hallucinated filters and mis-scoped queries.
- Built a **self-healing repair pipeline** triggered by **MongoDB Atlas change streams**: on schema changes, the frontier model rewrites affected skills & verifies stored answers before re-promoting, fully autonomously.

Self-Supervised Representation Learning for Active Matter Simulations | Github Link

April 2026

- Engineered a **fault-tolerant HPC training pipeline** on A100 GPUs with SIGUSR1-based preemption handling and step-level checkpointing to benchmark 11 SSL variants across 5 method families (**Conv-JEPA**, **3D ViT**, **ViT-JEPA**, **VideoMAE**) over 52GB of spatiotemporal physics simulation data; achieving a Linear Probing MSE of **0.011**.

Pico-LLM: Hardware-Efficient Transformer with DPO Alignment | Github Link

November 2025

- Built a Transformer-based LLM from scratch with **RMSNorm**, **Top-p sampling**, and **KV Cache**, benchmarking attention mechanisms from **Causal Self-Attention** to **Gated Linear Attention (GLA)**.
- Architected a complete **Direct Preference Optimization (DPO)** pipeline in **PyTorch** using a policy/reference model pair to align autoregressive generation with human preference data.

Microstructure Direct Estimation using a 2D U-Net | Github Link

March 2026

- Trained a **2D U-Net** in **PyTorch** to reconstruct **FA** and **MD** diffusion MRI microstructure maps from only 12 gradient directions, achieving a **13x acquisition speedup** over a full 160-direction protocol.
- Designed a **masked composite loss** with **Adam** optimization and **ReduceLROnPlateau** scheduling, reducing FA MAE from 0.0534 → **0.0354** and MD MAE from 0.0251 → **0.0078** across 100 epochs on the Stanford HARDI dataset.

ACHIEVEMENTS & CERTIFICATIONS

Winner, AI Builder Day Hackathon

July 2026

GDG Brooklyn × NYU Tandon — Voice-First Accessibility Navigation Assistant

Certificate

Winner, Nutanix Hackathon X (Surprise Me Category)

March 2024

Project Avalon — AI-driven chat interface for v4 VMM APIs

PUBLICATIONS

Optimization-Based Diabetes Prediction Using CNN & Bi-Directional LSTM

Link: <https://www.mdpi.com/2076-3417/12/8/3989>

Automatic Vehicle Identification and Classification Using the YOLOv3 for Toll System

Link: <https://www.mdpi.com/2071-1050/14/15/9163>